EFY Note
The source code of this project is available for free download at *www.electronicsforu.com*

account, and then create the OpenAI API as shown in Fig. 4.

Next, you need to install the library in the Arduino IDE according to the TFT display you are using, followed by the PS2 keyboard library.

In the code, after including the libraries, you need to configure the Wi-Fi SSID and the password of your Wi-Fi router and network. After that, configure the API key in the code that you have created earlier on the OpenAI website.

Fig. 5 shows a snippet of source code configuration for the Wi-Fi SSID and password. In the source code, you need to configure the OpenAI model you want to use. Fig. 6 shows a snippet of source code configuration for the OpenAI model. After completing all configurations in the source codes, upload the full source code by selecting the right COM port and the name of the board model.

Construction and testing

After uploading the source code and assembling the terminal, it is ready to be tested, as shown in Fig. 7.

The delays used inside the loops are very specific. You may change them, but start with these values, and then, once you have a handle on your responses, you may adjust them. You may begin with simple questions like: “translate ‘I love you’ into Bengali,” “translate ‘I love you’ into Chinese,” and “translate ‘I love you’ into Japanese.” ChatGPT meticulously produces the English scripts of the translated versions on the TFT screen.

Then, you may raise the level of questions to tasks like: “create an Arduino code for blinking,” “MicroPython code for blinking,” and “MicroPython code for text to speech.” Please note that ‘create an’ is optional; you may drop it as ChatGPT is very intelligent.

The highest level of questioning that was used with this prototype was to “write 5 sentences about Elektor magazine,” “write 5 sentences about NTPC,” and “write 5 sentences about BHEL.” For all the tests, ChatGPT performed exceptionally well.

This ESP32 ChatGPT is only a proof of concept to show that with a keyboard and an internet connection, AI can be used at the ESP32 MCU level as well. In the next development process, the keyboard may be replaced with speech-to-text interface or ChatGPT voice interactive solutions. EFY


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